

Node Types

Cache	Caching station
ClassSwitch	Class switching
Delay	Infinite server
Fork	Forks jobs
Join	Joins jobs
Logger	Trace logging
Place	Petri net place
Queue	Queueing station
Router	Routing node
Sink	Job departures (open)
Source	Job arrivals (open)
Transition	Petri net transition

Job Classes

```
ClosedClass(model, 'name', N, refS-
tat)
OpenClass(model, 'name')
SelfLoopingClass(model, 'name', N,
ref)
DisabledClass(model, 'name')
OpenSignal, ClosedSignal (catastrophes)
```

Distributions

APH(α , T)	Acyclic PH
Cox2(μ , ϕ)	Two-phase Coxian
Coxian(μ , ϕ)	Coxian
Det(v)	Deterministic
Disabled	None
Erlang(λ , k)	Erlang-k
Exp(λ)	Exponential
Gamma(α , β)	Gamma
HyperExp(p, λ_1 , λ_2)	Hyper-exp
Immediate	Zero service time
Lognormal(μ , σ)	Log-normal
Normal(μ , σ)	Normal
Pareto(α , k)	Pareto
PH(α , T)	Phase-type
Uniform(a, b)	Uniform
Weibull(α , β)	Weibull
Discrete	
Bernoulli(p)	Bernoulli
Binomial(p, n)	Binomial
DiscreteUniform(a, b)	Discr. uniform
Geometric(p)	Geometric
Poisson(λ)	Poisson
Zipf(s, n)	Popularity

Correlated / trace-driven

BMAP(D0...Dk)	Batch MAP
NHPP(bp, λ , cyclic)	Non-homog. Poisson
DMAP(D0, D1)	Discrete MAP
EmpiricalCDF(F)	Empirical CDF
MAP(D0, D1)	MAP
MarkedMAP	Marked MAP
ME(α , T)	Matrix-exp.
CME(mean, order)	Concentr. matrix-exp.
MMPP2(λ_1 , λ_2 , σ_1 , σ_2)	MMPP(2)
RAP(H0, H1)	Rational AP
Replayer(trace)	Trace replay

Scheduling Strategies

DPS	Discriminatory PS
EDD	Earliest Due Date
EDF	Earliest Deadline First
EXT	External (source)
FB	Feedback (LAS)
FCFS	First-Come First-Served
FCFSPI	FCFS Preempt. Resume-idle
FCFSR	FCFS Preemptive Resume
FSP	Fair Sojourn Protocol
GPS	Generalized PS
HOL	Head-of-Line Priority
INF	Infinite Servers
LCFS	Last-Come First-Served
LCFSPI	LCFS Preempt. Resume-idle
LCFSR	LCFS Preemptive Resume
LEPT	Longest Expected PT
LJF	Longest Job First
LPS	Limited PS
LRPT	Longest Remaining PT
OI	Order Independent
PAS	Pass and Swap
POLLING	Polling (switchover)
PS	Processor Sharing
PSJF	Preemptive SJF
SEPT	Shortest Expected PT
SETF	Shortest Elapsed Time First
SIRO	Random
SJF	Shortest Job First
SRPT	Shortest Remaining PT

Solvers

- AUTO Auto-select best
- BA Bound analysis (closed)
- CTMC CTMC (exact, small)
- ENV Random environments
- FLD Fluid/mean-field ODEs
- LDDES Discrete Event Sim
- LN Layered networks
- MAM Matrix Analytic Methods
- MVA Mean Value Analysis
- NC Normalizing Constant Methods
- SSA Stochastic simulation

Wrappers

- JMT JMT simulator
- LQNS LQNS interface
- QNS LQNS product-form solver

Solver Options

```
MVA(model, method='exact')
JMT(model, seed=123,
samples=10000)
SSA(model, samples=5000,
method='para')
LDES(model, MVA(model)) # warm start
```

Output Metrics

- ArvR Arrival rate
- QLen Mean queue length
- ResidT Residence time (total across visits)
- RespT Response time (per visit)
- Tput Throughput
- Util Server utilization
- QLenVar Queue-length variance
- QLenSCV Queue-length squared coeff. of variation
- QLenSkew Queue-length skewness
- RespTVar Response-time variance (FCFS only)
- RespTSCV Response-time SCV (FCFS only)
- RespTSkew Response-time skewness (FCFS only)

Moments (product form)

```
solver.get_moment_table()
# per class; order 2 by default
solver.get_moment_table(3)
# 1=means 2=var 3=skew
solver.get_moment_chain_table(3)
# per chain
solver.get_moment_station_table(3)
# per station total
solver.get_moment_station_table(3,'lin')
# approx., for large N
```

order is a set: scalar k = up to k ;
a vector, e.g. [2, 3], = exactly those.
RespT* moments are FCFS-only
(NaN elsewhere); exact methods cost
 $\prod_r (N_r + 1)$, use 'lin' beyond that.

Analysis Methods

```
solver.avg_table()
solver.get_avg_qlen(), get_avg_util()
solver.get_avg_respt(), get_avg_tput()
solver.get_avg_sys_respt()
solver.get_avg_sys_tput()
solver.get_avg_qlen_chain()
```

Table Methods

- avg_table per station
- get_avg_chain_table per chain
- get_avg_node_table per node
- get_avg_node_chain_table node $\times$ chain
- get_avg_sys_table system totals

Routing

```
model.link(Network.serial_routing(
n1,n2,n3))
n1.set_prob_routing(jobclass, n2, 0.7)
n1.set_routing(jobclass,
RoutingStrategy.RROBIN)
```

Multi-Server & Finite Capacity

```
queue.set_number_of_servers(k)
queue.set_capacity(bufferSize)
```

Class Switching

```
cs = ClassSwitch(model, 'CS')
P = [[0.3, 0.7], [0.5, 0.5]]
cs.set_class_switching_matrix(P)
```

Fork-Join Networks

```
fork = Fork(model, 'Fork')
join = Join(model, 'Join', fork)
```

Layered Networks

```
lqn = LayeredNetwork('name')
P = Processor(lqn,'P',1,
SchedStrategy.PS)
T = Task(lqn,'T',1,
SchedStrategy.REF).on(P)
E = Entry(lqn,'E').on(T)
A = Activity(lqn,'A').on(T) \
.bound_to(E).set_host_demand(Exp(1))
```

Caching

```
cache = Cache(model,'Cache',
nItems,cap,ReplacementStrategy.LRU)
cache.set_read(jobclass, Zipf(1.4, nItems))
# cache results
MVA(model).get_avg_cache_table() # hit/miss
MVA(model).get_avg_item_table() # per-item
```

Replacement Strategy

- CLIMB Climb
- FIFO First-in first-out
- HLRU h-LRU
- LRU Least recently used
- QLRU q-LRU
- RR Random replacement
- SFIFO Static FIFO

Model Import/Export

```
model.jsimg_view() # Open in JMT
save_model(model,'model.json')
model = load_model('model.json')
```

Routing Strategies

- JSQ Join shortest queue
- KCHOICES Power of k choices
- PROB Probabilistic routing
- RAND Random selection
- RROBIN Round-robin
- WRROBIN Weighted round-robin

Stochastic Petri Nets

```
p = Place(model, 'P1')
t = Transition(model, 'T1')
m = t.add_mode('Model')
t.set_number_of_servers(m, float('inf'))
t.set_distribution(m, Exp(4))
t.set_enabling_conditions(m, jobclass, p, 1)
t.set_inhibiting_conditions(m, jobclass, p, 3)
t.set_firing_outcome(m, jobclass, sink, 1)
```

Markov Chains (solved directly)

```
ctmc = MarkovProcess(Q)
pi = SolverCTMC(ctmc).get_prob_sys()
dtmc = MarkovChain(P)
piD = SolverCTMC(dtmc).get_prob_sys()
piT = ctmc.transient(None, 10.0)
jump = ctmc.toEmbedded()
```

Random Environments

```
env = Environment('R', 2)
env.add_stage(0,'Fast','operational',fastModel)
env.add_stage(1,'Slow','degraded',slowModel)
env.add_transition(0,1,Exp(0.5))
env.add_transition(1,0,Exp(1.0))
ENV(env, lambda m: FLD(m)).avg_table()
```

Finite Capacity Regions

```
fcr = model.add_region([queue1, queue2])
fcr.set_global_max_jobs(10)
fcr.set_class_max_jobs(class1, 5)
fcr.set_drop_rule(class1, DropStrategy.DROP)
```

Load-Dependent Service

```
queue.set_load_dependence(np.minimum(np.arange(1,N+1), c))
queue.set_class_dependence(lambda ni: min(sum(ni), c), c)
```

Impatience & Retrials

```
queue.set_patience(jobclass,
ImpatienceType.RENEGING, Exp(0.1))
queue.set_balking(jobclass, strategy, thresholds)
queue.set_retryal(jobclass, Exp(2), maxAttempts)
```

State Probabilities

```
solver.get_prob(node) # marginal
solver.get_prob_aggr(node) # aggregate
solver.get_prob_sys() # joint
NC(m).get_prob_sys_marg(n) # totals
solver.get_generator() # CTMC infin.
solver.get_state_space()
```

Transient & Distributions

```
solver.get_tran_avg() # transient means
solver.get_cdf_respt() # response-time CDF
model.set_reward('Cost', lambda state: ...)
solver.get_avg_reward()
```